The universe is granular, not continuous. With infinitely small points, it would be impossible ever to construct extension – as in Democritus's argument, reported by Aristotle and mentioned above. Therefore, the extension of the string must be formed by a *finite* number of finite objects with *finite* size. The string *cannot* be cut *as many times as we want*; matter is not continuous, it is made of individual 'atoms' of a finite size.

Whether this abstract argument is correct or not, its conclusion – as we know today – contains a great deal of truth. Matter does indeed have an atomic structure. If I divide a drop of water in two, I obtain two drops of water. I can divide each one of these two drops again, and so on. But I cannot continue to infinity. At a certain point I have only one molecule, and I have finished. No drops of water exist smaller than a single molecule of water.

How do we know this today? Evidence has accumulated over centuries, much of it from chemistry. Chemical substances are made up of combinations of a few elements and are formed by proportions (of weight) given by whole numbers. Chemists have constructed a way of thinking about substances as composed of molecules made up of fixed combinations of atoms. Water, for example – H_2O – is composed of two parts hydrogen and one part oxygen.

But these are only clues. Still at the beginning of the last century numerous scientists and philosophers did not consider the atomic hypothesis to be credible. Among them was the renowned physicist and philosopher Ernst Mach, whose ideas on space would come to have great importance for Einstein. At the end of a lecture by Ludwig Boltzmann at the Imperial Academy of Science in Vienna, Mach publicly declared, 'I do not believe that atoms exist!' This was in 1897. Many, like Mach, understood chemical notation only as a conventional method of summarizing laws of chemical reactions – not as evidence that there actually were molecules of water composed of two atoms of hydrogen and one of oxygen. You can't see atoms, they would say. Atoms will never be seen, they would say. And then, they asked, how big would an atom be? Democritus could never measure the size of his atoms …

But somebody else could. The definitive proof of the 'atomic hypothesis' had to wait until 1905. It was found by a rebellious twenty-five-year-old, who had studied physics but had not been able to find employment as a scientist and was making ends meet by working as an employee in the patent office in Berne. I will speak a lot about this young man in the rest of this book, and about the three articles he sent to the most prestigious physics journal of the time, the *Annalen der Physik*. The first of these articles contained the definitive proof that atoms exist, and calculated their dimensions, solving the problem posed by Leucippus and Democritus twenty-three centuries earlier.